Multi-Model AI Agent for Automated Health Diagnostics

A Modular AI System for Automated Blood Report Interpretation

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Infosys Springboard Internship Program

Problem Statement & Objective

Problem

Blood reports are hard to interpret for non-clinical users; reports arrive in heterogeneous formats (PDF, image scans, JSON). Manual review requires medical expertise and is time-consuming.

Objective

Build an automated pipeline to interpret blood reports, detect clinical patterns, quantify risk, and produce structured, evidence-aligned recommendations for users and clinicians.

Data Sources & OCR

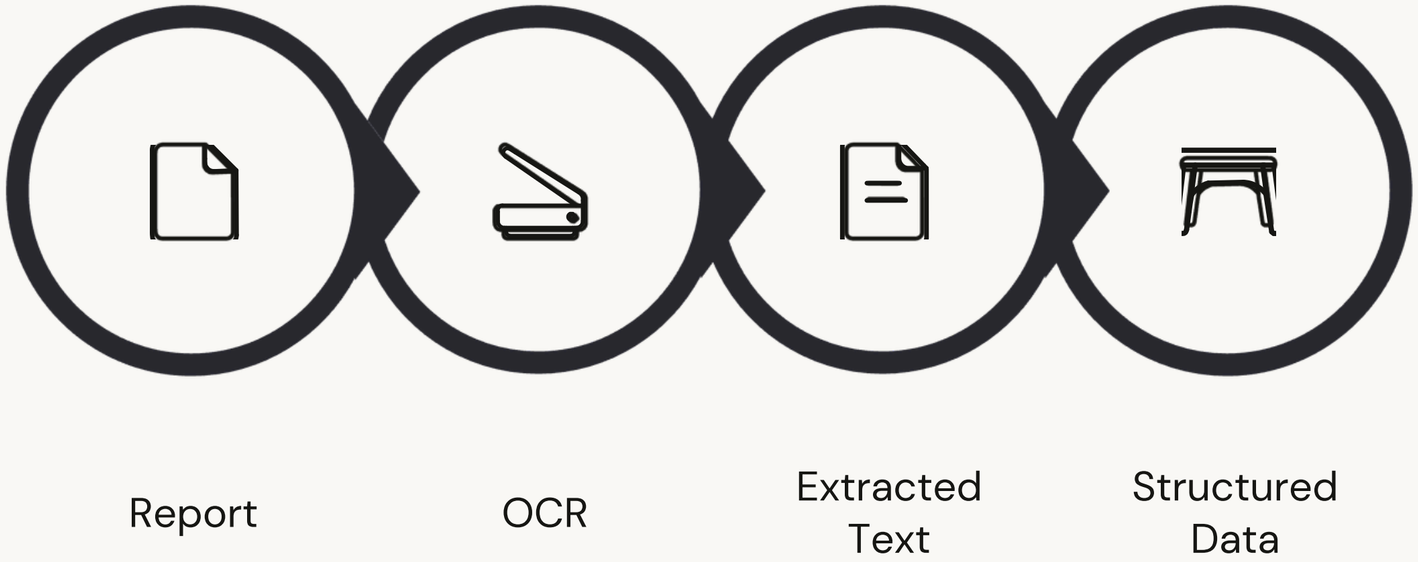
Input Sources

- Laboratory reports (sample development corpus)
- Formats: JSON exports, machine PDFs, scanned images

Ensuring broad coverage of real-world report formats improves model robustness.

OCR: Role & Value

Optical Character Recognition converts image/PDF text to machine-readable text, enabling downstream parsing, normalization, and structured extraction.



OCR accuracy is a gating factor; preprocessing (de-skew, enhance) improves extraction rates.

Data Extraction & Processing

01

Regex-Based Extraction

Pattern-driven extraction locates parameter names, numeric values, and units across noisy layouts.

02

Parameter Identification

Assign canonical parameter IDs (e.g., Hemoglobin → HGB) to support downstream logic and mapping across lab vocabularies.

03

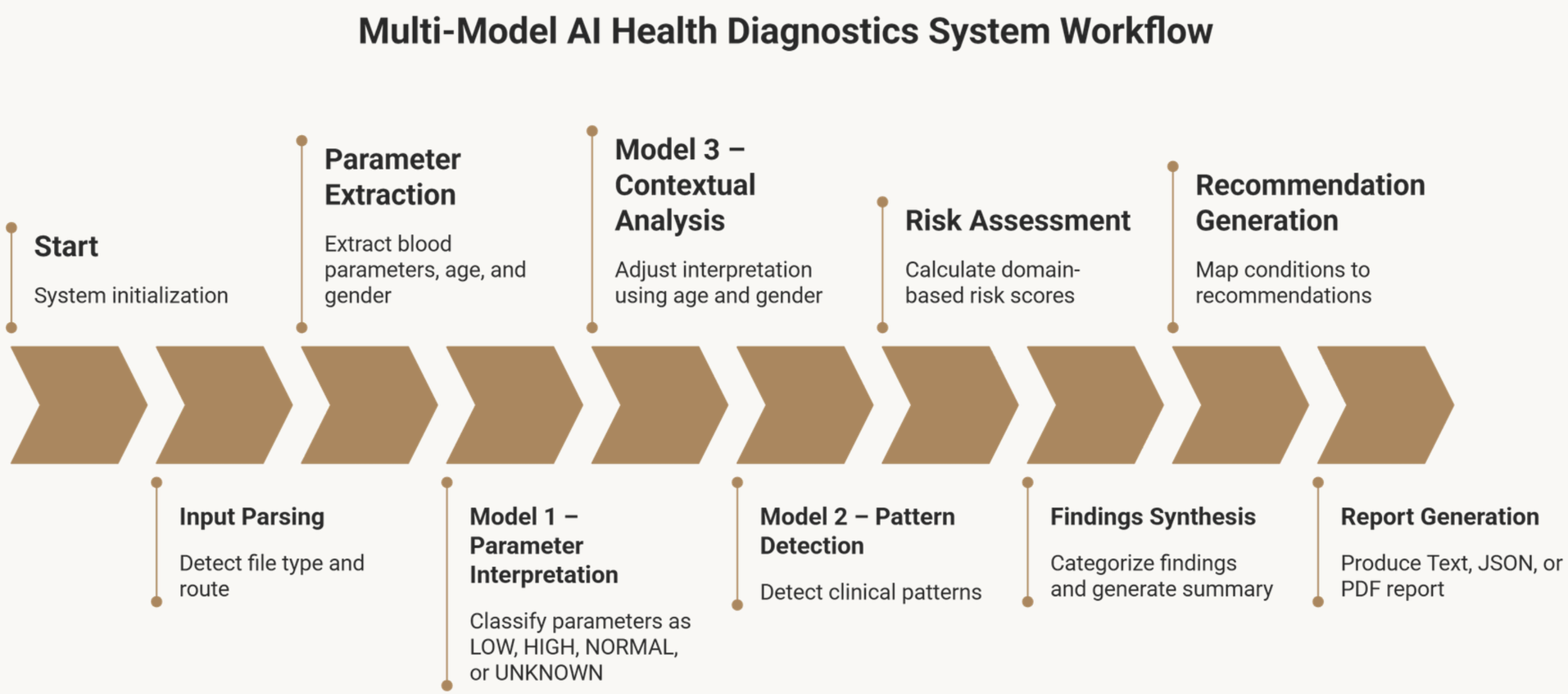
Validation & Standardization

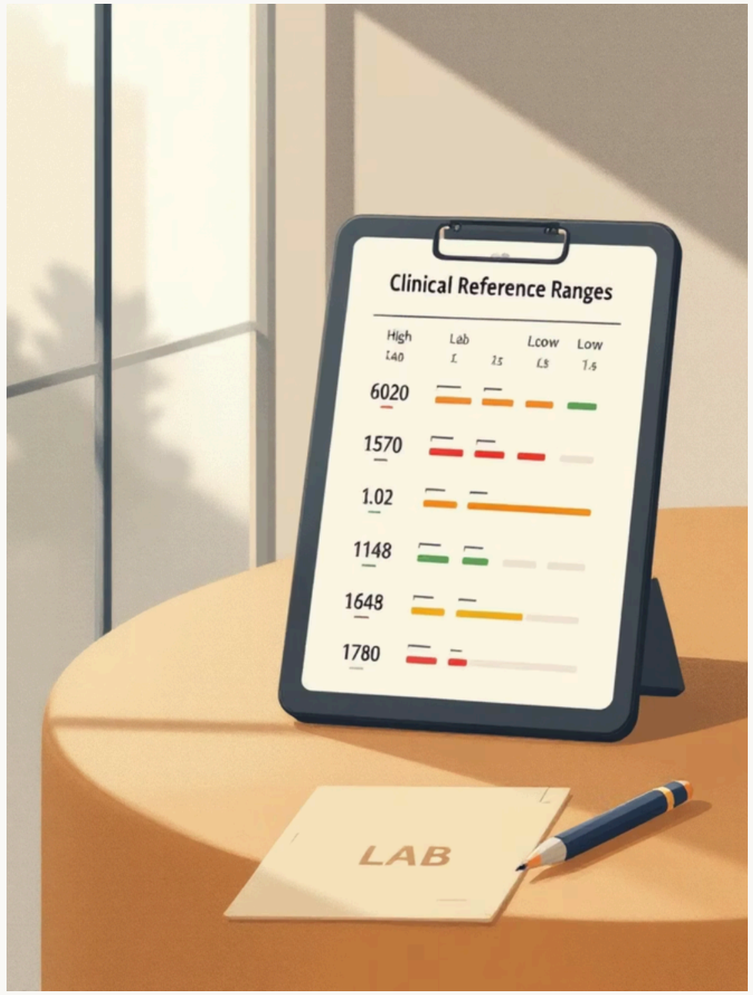
Unit normalization, value sanity checks, and handling of formatting variants (commas, ranges, flags) ensure consistent structured data.

Overall System Workflow

End-to-end pipeline: Ingestion → Extraction & Validation → Interpretation models → Pattern detection → Contextual analysis → Risk scoring → Recommendation generation → Final structured report.

Modular orchestration enables independent testing, versioning, and failover for each stage.

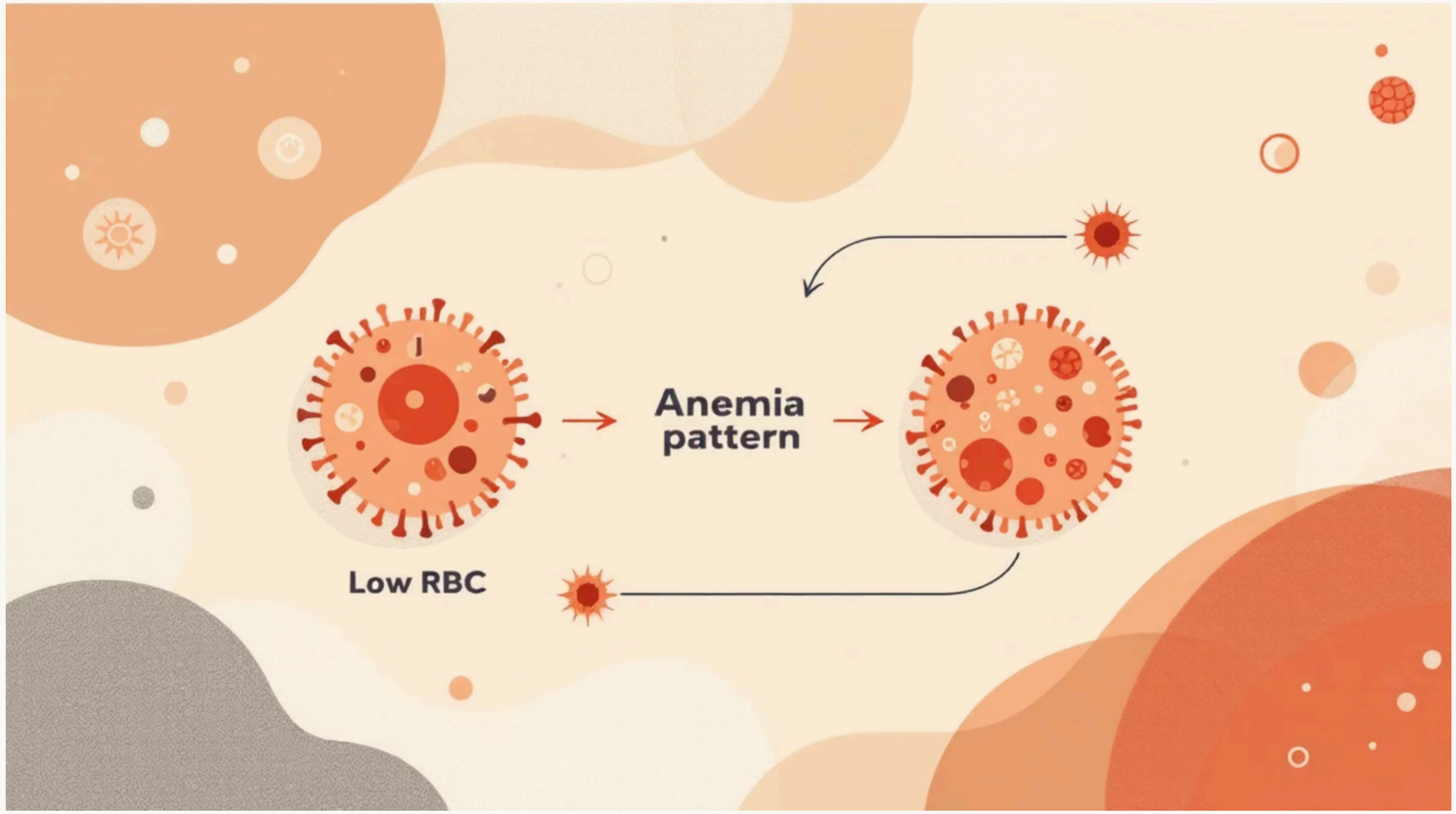




Model 1 — Parameter Interpretation

Model 1 maps each extracted value to medical reference ranges and classifies results as Low, Normal, High, or Borderline using rule-based thresholds informed by source lab metadata.

- Reference-range lookup (lab-specific when available)
- Age- and sex-aware baseline selection
- Deterministic rule logic forms the interpretation foundation



Model 2 & Model 3 — Pattern Detection and Contextual Analysis

Combined outputs feed a synthesis layer that balances per-parameter flags with multi-parameter patterns.

Pattern Detection

Identifies co-occurring abnormalities (e.g., Low HGB + Low RBC → anemia pattern) and assigns confidence scores to pattern matches.

Contextual Analysis

Adjusts interpretation using demographic context (age, sex) and historical values, refining thresholds and reducing false positives.

Risk Assessment & Synthesis

Scoring Mechanism

Weighted aggregation of model outputs produces a normalized 0–100 risk score representing clinical concern magnitude.

Risk Categories

Five-tier labels: Minimal, Low, Moderate, High, Critical — calibrated to clinical thresholds and configurable per deployment.

Synthesis Engine

Combines individual parameter interpretation, detected patterns, and contextual modifiers to produce a concise clinical summary.

Summary outputs include supporting evidence (which parameters contributed most) and confidence metrics for reviewer transparency.

Recommendation Engine & Orchestrator

Personalized Recommendations

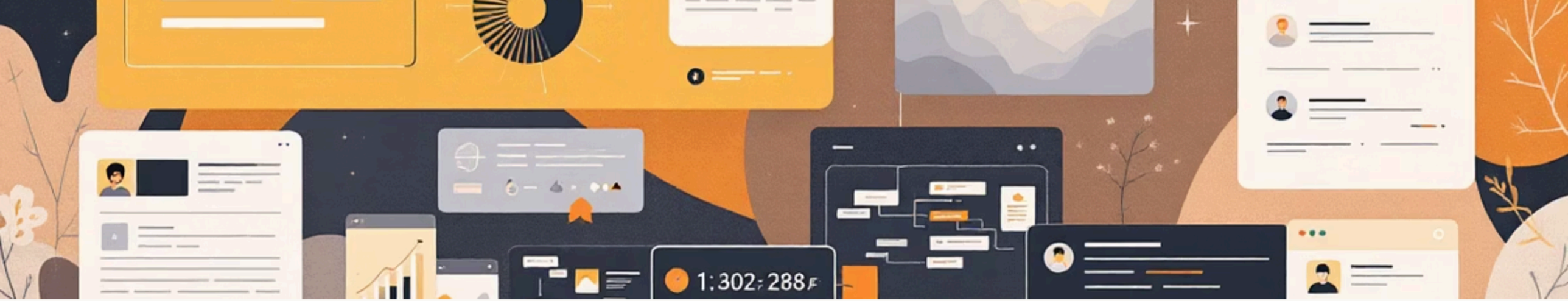
Dietary and lifestyle advice mapped directly to detected abnormalities (e.g., iron-rich diet for iron-deficiency pattern).

Follow-up Guidance

Suggested timelines for repeat testing, urgent referral flags, and red-flag escalation when risk is High/Critical.

Orchestration & Reliability

Multi-model orchestrator sequences model execution, manages retries, logs errors, and guarantees end-to-end reliability.



Technical Highlights & Conclusion

Core Components

OCR-based extraction • Regex-driven parsing • Context-aware medical logic • Confidence-based pattern detection

Architecture

Modular, testable pipelines with orchestrator and clear interfaces for model updates and clinical governance.

Interface & Deployment

Streamlit-based prototype UI; designed for scalability and integration with EHRs or telehealth portals.

Conclusion: An end-to-end, extensible system that converts raw lab reports into actionable clinical insights — improving accessibility, reproducibility, and timely decision support.

Thank You!!!